

Anisha Ramesh

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EDUCATION

University of California San Diego

San Diego, CA

Bachelor of Science in Computer Engineering; GPA: 3.887/4.0

Sep 2024 – June 2028

- Provost Honors: Fall '24, Winter '25, Spring '25, Fall '25
- Leadership: Board Member of Triton Engineering Student Council, Marketing Director of Student-Initiated Programs, Instructional Assistant for CSE 8A/6R: Introduction to Computer Science and Object-Oriented Programming in Python (Fall 2025 and Winter 2026), CS Team Lead for Envision Conference
- Coursework: Computational Problem Solving, Discrete Mathematics, Basic Data Structures & Object-Oriented Design, Systems Programming & Software Tools, Computer Organization, Algorithms & Systems Analysis

EXPERIENCE

Instructional Assistant - CSE 8A/6R

Sep 2025 – Present

Computer Science and Engineering Department at UCSD

San Diego, CA

- Tutored 30+ students through weekly labs and study groups
- Graded 100+ homework assignments and assisted in designing the lab write-ups
- Collaborated with the other tutors and teaching assistants to design the grading rubric, class logistics and class policies
- Provided one-on-one assistance to students through personalized explanation videos and individual tutor hours

Software Engineer Intern

June 2025 – Sep 2025

San Diego Supercomputer Center

San Diego, CA

- Contributed to development of centralized club management platform with 6-member team to increase student engagement and reduce administrative workload
- Built scalable full-stack features with React, Node.js, and MongoDB to improve platform performance
- Accelerated product delivery using Agile workflows and Sprint Reviews with thorough SDLC training
- Worked collaboratively with my team through daily sync meetings, merging features and debugging sessions
- Project Documentation: sdsc-ucsd.atlassian.net
- Source Code: <https://github.com>

PROJECTS

ML-Based Music Classifier

Python, Scikit-Learn, Librosa, Pandas

2025

- * Developed end-to-end machine learning system to classify music genres using audio signal processing techniques
- * Built feature extraction pipeline with Librosa and implemented Logistic Regression, achieving 69% accuracy on 10-genre classification
- * Conducted hyperparameter tuning analysis revealing near-optimal initial model conditions through effective preprocessing and feature selection
- * Source Code: <https://github.com/anisharamesh21/MusicClassifier>

Huffman Coding Text Compression Tool

Python, Data Structures, Algorithm Design

2025

- * Implemented Huffman coding algorithm to compress text files, reducing file size by 47% compared to standard ASCII encoding
- * Built custom Node class and generated efficient variable-length codes for optimal compression performance
- * Analyzed and visualized character frequencies using Pandas, Matplotlib, and Seaborn for comprehensive data insights
- * Source Code: <https://github.com/anisharamesh21/Huffman-Coding-Project>

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++ JavaScript, TypeScript, Assembly Language

Web Technologies: HTML, CSS, React, Node.js, MongoDB

Data & Analytics: NumPy, Pandas, SciPy, Matplotlib, Seaborn, Scikit-Learn, Librosa, pdfplumber, gTTS, pyttsx3, PyTorch

Developer Tools: Git, GitHub/GitLab, Unix Commands, GDB, Valgrind, Makefile

Core Competencies: Machine Learning, Data Structures, Object-Oriented Programming, Full-Stack Development